Lecture exercise branching

This notebook was generated in solutions mode.

Exercise: Positive, Negative, or Zero

Write a program that determines whether a given number is positive, negative, or zero.

Use the number `num = -5` and store the result in variable `result` ("positive", "negative", or "zero").

```
num = -5
if num > 0:
    result = "positive"
elif num < 0:
    result = "negative"
else:
    result = "zero"
```

```
assert result == "negative"
```

Python Tutor Visualization

The following cells contain code to generate links to Python Tutor for visualizing this exercise. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML(f'<a href="https://pythontutor
```

Exercise: Even or Odd Number

Write a program that determines whether a given integer is even or odd.

Use the number `num = 7` and store the result in variable `result` ("even" or "odd").

```
num = 7
if num % 2 == 0:
    result = "even"
else:
    result = "odd"
```

```
assert result == "odd"
```

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```
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```

Exercise: Grade Evaluator

Write a program that converts a numerical score (0-100) to a letter grade:

- 90-100: A
- 80-89: B
- 70-79: C
- 60-69: D
- 0-59: F

Use the score `score = 85` and store the letter grade in variable `result`.

```
assert result == "B"
```

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```

Exercise: Divisibility Checker

Write a program that checks if a given number is divisible by both 3 and 5.

Use the number `num = 30` and store the result in variable `result` ("divisible by both" or "not divisible by both").

```
num = 30
if num % 3 == 0 and num % 5 == 0:
    result = "divisible by both"
else:
    result = "not divisible by both"
```

```
assert result == "divisible by both"
```

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```
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```

Exercise: Leap Year Checker

Write a program that determines whether a given year is a leap year or not.

A leap year is:

- Divisible by 4 but not by 100, OR
- Divisible by 400

Use the year `year = 2024` and store the result in variable `result` ("leap year" or "not leap year").

```
year = 2024
if (year % 4 == 0 and year % 100 != 0) or year % 400 == 0:
    result = "leap year"
else:
    result = "not leap year"
```

```
assert result == "leap year"
```

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```
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```

Exercise: Vowel or Consonant

Write a program that determines whether a given letter is a vowel or consonant.

Use the letter `letter = "A"` and store the result in variable `result` ("vowel" or "consonant").

```
letter = "A"
if letter in "aeiouAEIOU":
    result = "vowel"
else:
    result = "consonant"
```

```
assert result == "vowel"
```

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```

Exercise: Password Validator

Write a program that validates a password. If the password is "password123", grant access, otherwise deny access.

Use the password `password = "password123"` and store the result in variable `result` ("Access granted" or "Access denied").

```
password = "password123"
if password == "password123":
    result = "Access granted"
else:
    result = "Access denied"
```

```
assert result == "Access granted"
```

Python Tutor Visualization

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```
from IPython.display import HTML;import urllib.parse as u;HTML(f'<a href="https://pythontutor
```

Exercise: Largest of Three Numbers

Write a program that finds the largest of three given numbers.

Use the numbers `num1 = 15`, `num2 = 8`, and `num3 = 22`. Store the largest number in variable `result`.

```
num1 = 15
num2 = 8
num3 = 22
if num1 >= num2 and num1 >= num3:
    result = num1
elif num2 >= num1 and num2 >= num3:
    result = num2
else:
    result = num3
```

```
assert result == 22
```

Python Tutor Visualization

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```
from IPython.display import HTML;import urllib.parse as u;HTML(f'<a href="https://pythontutor
```

Exercise: Day of the Week

Write a program that converts a number (1-7) to the corresponding day of the week:

- 1: Monday
- 2: Tuesday
- 3: Wednesday
- 4: Thursday
- 5: Friday
- 6: Saturday
- 7: Sunday
- Any other number: "Invalid number"

Use the number `day = 5` and store the day name in variable `result`.

```
day = 5
if day == 1:
    result = "Monday"
```

```python
elif day == 2:
    result = "Tuesday"
elif day == 3:
    result = "Wednesday"
elif day == 4:
    result = "Thursday"
elif day == 5:
    result = "Friday"
elif day == 6:
    result = "Saturday"
elif day == 7:
    result = "Sunday"
else:
    result = "Invalid number"
```

```python
assert result == "Friday"
```

```python
# Test with invalid number
test_day = 8
if test_day == 1:
    test_result = "Monday"
elif test_day == 2:
    test_result = "Tuesday"
elif test_day == 3:
    test_result = "Wednesday"
elif test_day == 4:
    test_result = "Thursday"
elif test_day == 5:
    test_result = "Friday"
```

```
elif test_day == 6:
    test_result = "Saturday"
elif test_day == 7:
    test_result = "Sunday"
else:
    test_result = "Invalid number"
assert test_result == "Invalid number"
```

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```
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```

Exercise: Simple Calculator

Write a program that performs basic arithmetic operations (+, -, *, /) on two numbers based on the given operation.

Use `num1 = 10.0`, `num2 = 3.0`, and `operation = "+"`. Store the result in variable `result`. Handle division by zero with an error message.

```
num1 = 10.0
num2 = 3.0
```

```python
operation = "+"
if operation == "+":
    result = num1 + num2
elif operation == "-":
    result = num1 - num2
elif operation == "*":
    result = num1 * num2
elif operation == "/":
    if num2 != 0:
        result = num1 / num2
    else:
        result = "Error: Cannot divide by zero"
else:
    result = "Invalid operation"
```

```python
assert result == 13.0
```

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```python
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```

Exercise: Triangle Type Classifier

Write a program that determines the type of triangle based on the lengths of its three sides:

- Equilateral: All sides are equal
- Isosceles: Two sides are equal
- Scalene: No sides are equal

Use `side1 = 5.0`, `side2 = 5.0`, and `side3 = 8.0`. Store the triangle type in variable `result`.

```
side1 = 5.0
side2 = 5.0
side3 = 8.0
if side1 == side2 == side3:
    result = "equilateral"
elif side1 == side2 or side2 == side3 or side1 == side3:
    result = "isosceles"
else:
    result = "scalene"
```

```
assert result == "isosceles"
```

Python Tutor Visualization

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```python
from IPython.display import HTML;import urllib.parse as u;HTML(f'<a href="https://pythontutor
```

Exercise: Maximum of Three Numbers

Write a program that finds the maximum (largest) of three given numbers.

Use the numbers `num1 = 12.5`, `num2 = 8.3`, and `num3 = 15.7`. Store the maximum number in variable `result`.

```python
num1 = 12.5
num2 = 8.3
num3 = 15.7
if num1 >= num2 and num1 >= num3:
    result = num1
elif num2 >= num1 and num2 >= num3:
    result = num2
else:
    result = num3
```

```python
assert result == 15.7
```

```python
# Test with different maximum
test_num1, test_num2, test_num3 = 25.0, 18.5, 20.2
if test_num1 >= test_num2 and test_num1 >= test_num3:
    test_result = test_num1
```